

Are explanations less reliable when the model is wrong? Post-hoc explanation quality on misclassified instances

¿Son menos fiables las explicaciones cuando el modelo se equivoca? Calidad de las explicaciones post hoc en instancias mal clasificadas

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Abstract

Explanations of machine-learning predictions are consulted most when a prediction is in doubt, yet explainers are evaluated on averages over all instances. We ask whether post-hoc explanations are less reliable when the model is wrong. Using an existing benchmark on the UCI Adult data, we compare fidelity, stability and faithfulness gap between correctly and incorrectly classified instances for SHAP, LIME, Anchors and DiCE across five model families, with the run as the unit of analysis (265 runs, 122,529 explained instances), Wilcoxon signed-rank tests with Holm correction, a control for the prediction margin, and an external check on German Credit. Explanations were less stable on misclassified instances for all four explainers, and the deficit was large for SHAP (median difference -0.076 , $d_z = -1.36$) and DiCE (-0.210 , $d_z = -2.15$). The prediction margin explained most of the DiCE deficit but not the SHAP deficit. Holding the margin fixed, the features that SHAP, LIME and Anchors ranked first carried less of the prediction on errors. Fidelity changed little and in opposite directions across explainers. The pattern depended on the model family, and on German Credit it held for Anchors but not for SHAP stability. Quality reported as an average can therefore overstate how far an explanation of a wrong prediction should be trusted; audits should report explanation quality separately for correct and incorrect predictions.

Keywords: explainable artificial intelligence; post-hoc explanations; misclassification; explanation reliability; SHAP; LIME.

Resumen

Las explicaciones de las predicciones de aprendizaje automático se consultan sobre todo cuando una predicción es dudosa, pero los explicadores se evalúan con promedios sobre todas las instancias. Nos preguntamos si las explicaciones post hoc son menos fiables cuando el modelo se equivoca. Con un

banco de pruebas existente sobre UCI Adult, comparamos fidelidad, estabilidad y brecha de borrado entre instancias bien y mal clasificadas para SHAP, LIME, Anchors y DiCE en cinco familias de modelos, con la ejecución como unidad de análisis (265 ejecuciones, 122,529 instancias), pruebas de Wilcoxon con corrección de Holm, un control del margen de predicción y una verificación externa con German Credit. Las explicaciones fueron menos estables en las instancias mal clasificadas para los cuatro explicadores, y el déficit fue grande para SHAP (diferencia mediana -0.076 , $d_z = -1.36$) y DiCE (-0.210 , $d_z = -2.15$). El margen de predicción explicó la mayor parte del déficit de DiCE, pero no el de SHAP. Con el margen fijo, las variables que SHAP, LIME y Anchors situaron en primer lugar sostuvieron una parte menor de la predicción en los errores. El patrón dependió de la familia de modelos y, en German Credit, se mantuvo para Anchors pero no para la estabilidad de SHAP. La calidad promedio puede exagerar la confianza que merece la explicación de una predicción errónea; las auditorías deberían informarla por separado para predicciones correctas e incorrectas.

Palabras clave: inteligencia artificial explicable; explicaciones post hoc; errores de clasificación; fiabilidad de las explicaciones; SHAP; LIME.

1 Introduction

Post-hoc explainers such as SHAP [1], LIME [2], Anchors [3] and DiCE [4] are the usual way to make the predictions of an opaque classifier inspectable [5]. Their quality is assessed with proxy metrics: whether the features an explanation ranks as important are those that move the prediction (fidelity and deletion tests [6–8]), and whether similar inputs receive similar explanations (stability [9]). These metrics are reported as averages over a sample of instances, and benchmarks compare explainers by those averages [10].

An average hides how quality is distributed across instances, and one split matters more than most: whether the model’s prediction was right. An explanation is rarely consulted to confirm an obviously correct decision. It is consulted when a decision is contested, when an auditor examines a failure, or when a user must decide whether to accept a recommendation. In those situations the prediction is disproportionately likely to be wrong. User studies show that explanations can raise acceptance of a model’s recommendation whether or not the recommendation is correct [11], and explanations can be unreliable in ways that their appearance does not reveal [12, 13]. If explanations are also less faithful or less stable on misclassified instances, an explanation can look convincing exactly where it deserves the least trust [14, 15].

Whether explanation quality differs between correct and incorrect predictions is an empirical question, and the usual reporting cannot answer it. We answer it with a pre-specified re-analysis of an existing benchmark that explained, under one protocol, balanced samples of true positives, true negatives, false positives and false negatives for four explainers and five model families. We ask

four questions:

- **RQ1.** Does explanation quality differ between correctly and incorrectly classified instances, for each explainer?
- **RQ2.** Does the difference depend on the model family?
- **RQ3.** Among errors, do false positives and false negatives differ?
- **RQ4.** Is the difference explained by proximity to the decision boundary? Errors tend to lie near it, and some metrics may depend on the margin rather than on correctness itself.

The contribution is an instance-level answer, with the run as the unit of inference, a margin control, an external check on a second dataset, and a practical recommendation for how explanation audits should report quality.

2 Materials and methods

2.1 Benchmark and data

The data are the runs of an existing benchmark of post-hoc explainers on the UCI Adult income data [16, 17], released with its protocol and reproducibility contract in [10]. That study, and a later analysis of the same runs, compare explainers by their average quality over all instances. The present analysis is different and reports none of their results: it compares each explainer with itself, between correct and misclassified instances of the same run. Re-using the cohort rather than re-executing it keeps the explainers, models and instances fixed, so the only contrast is the correctness of the prediction.

Five classifiers, each trained once, were explained: logistic regression, random forest [18], gradient-boosted trees (XGBoost) [19], a support vector machine and a multilayer perceptron. Each run crossed one model family with one explainer, one random seed (42, 123, 456, 789 or 999) and one sampling intensity of $n = 50, 100$ or 200 instances per error quadrant. The seed fixed the partition from which instances were drawn; the models were trained on the partition of seed 42. Within a run, test instances were sampled separately from the four quadrants (TP, TN, FP, FN), so correct and misclassified instances are represented in nearly equal numbers by design.

Of the 299 run files, 265 entered the analysis. Anchors and DiCE produced no explanations in 18 and 7 runs respectively (failed or non-converged), and 9 SHAP runs had fewer than ten instances in one of the two groups. The analysis covers 122,529 explained instances; 278 instances without a recorded quadrant were excluded. Table 1 gives the counts per explainer.

Table 1: Runs and instances analysed per explainer (UCI Adult).

Explainer	Runs present	Runs analysed	Correct	Misclassified
SHAP	74	65	15,369	14,302
LIME	75	75	17,500	17,103
Anchors	75	57	12,992	12,775
DiCE	75	68	16,300	16,188

For the external check we used the German Credit data [20] from the same benchmark: random forest and XGBoost, explained by SHAP and Anchors, three seeds each (12 runs). These runs have 13 to 37 errors per quadrant. A third dataset in the benchmark, Breast Cancer, was excluded in advance because it has only one to four errors per quadrant per run.

2.2 Explainers and quality metrics

SHAP used TreeExplainer for the tree models and KernelExplainer otherwise; LIME used a tabular surrogate with ten features. Both return an attribution vector $e(x) \in \mathbb{R}^M$ over the M model features. Anchors returns a rule; it is represented as a binary vector marking the features in the rule. DiCE returns a counterfactual x' of the opposite class, represented as $|x - x'|$. Let $f(x)$ be the model’s predicted probability of the positive class, and let masking feature j replace its value with the training mean. The metrics, computed per instance by the benchmark [10], are:

Stability S [9]: the explainer is run on T copies of x perturbed with Gaussian noise, and S is the mean cosine similarity of the resulting vectors over all pairs:

$$S = \frac{2}{T(T-1)} \sum_{a < b} \cos(e^{(a)}, e^{(b)}). \quad (1)$$

Fidelity F , a single-feature form of faithfulness correlation [6]: the Pearson correlation, across features, between the magnitude of each attribution and the change in prediction when that feature alone is masked:

$$F = \rho \left(\{|e_j(x)|\}_{j=1}^M, \{|f(x) - f(\tilde{x}^{(j)})|\}_{j=1}^M \right). \quad (2)$$

Faithfulness gap G , a deletion test [7, 8]: the change in prediction when the $k = 5$ features with the largest attributions are masked together:

$$G = |f(x) - f(\tilde{x}^{(\text{top-}k)})|. \quad (3)$$

Sparsity, the share of features with a non-negligible attribution, was analysed as a secondary metric. Explanation time was summarised descriptively. For all three primary metrics, higher values

indicate a better explanation.

2.3 Statistical analysis

The analysis plan, with the questions, metrics, tests and corrections, was written and committed to the project repository before any result was computed; its two deviations are reported below. Instances are nested in runs, so instance-level tests would treat thousands of dependent explanations as independent. The run is therefore the unit of inference. For run r and metric Q ,

$$\Delta_r = \bar{Q}_{r,\text{mis}} - \bar{Q}_{r,\text{cor}}, \quad (4)$$

the mean over misclassified instances (FP and FN) minus the mean over correct instances (TP and TN). A negative Δ means lower quality on errors.

RQ1: for each explainer and primary metric (12 tests), a two-sided Wilcoxon signed-rank test of $\Delta = 0$ across runs [21], with Holm correction across the 12 tests [22] at $\alpha = 0.05$. We report the median Δ , the mean with a 95% confidence interval, and the standardised effect d_z (mean over standard deviation of Δ) [23]. An effect is called practically relevant when it is significant after correction, $|d_z| \geq 0.5$ and $|\text{median } \Delta| \geq 0.05$. As a robustness check, the analysis was repeated after averaging Δ over the three sampling intensities within each model and seed. **RQ2:** Δ by model family, with a Kruskal–Wallis test [24] per explainer and metric (Holm across 12). **RQ3:** as RQ1 with $\Delta_r = \bar{Q}_{r,\text{FP}} - \bar{Q}_{r,\text{FN}}$.

RQ4: misclassified instances lie closer to the decision boundary, and metrics that perturb or mask the input may behave differently there. The margin of an instance, $m = |\hat{p} - 0.5|$, was recomputed from the stored models. For each explainer and metric we fitted, at instance level,

$$Q_{ir} = \alpha_r + \beta \text{mis}_{ir} + \gamma m_{ir} + \varepsilon_{ir}, \quad (5)$$

with run fixed effects α_r and standard errors clustered by run [25], and compared β with its value when the margin is omitted. A difference is called explained by the margin when β shrinks by more than half and its confidence interval includes zero. The external check repeats RQ1 on German Credit descriptively: with six runs per explainer, no test can reach significance after correction.

Deviations from the plan. (1) For the random-forest runs executed first, the stored model reproduced only part of the recorded predictions, so their margin cannot be recomputed; RQ4 therefore uses the 223 of 274 runs whose recorded predictions the stored model reproduces (at least 99%), which leaves 9 of 60 random-forest runs. RQ1–RQ3 are unaffected, because their correct and misclassified labels were assigned by the model that actually ran. (2) For the SVM, $\hat{p}$ is the Platt-scaled probability [26], which disagrees with the predicted class for 404 of 22,426 instances; they were

kept.

3 Results

3.1 Correct versus misclassified instances (RQ1)

Table 2 and Figure 1 give the per-run differences. Stability was lower on misclassified instances for all four explainers after correction. The deficit was large for SHAP, lower in all 65 runs (median -0.076 , $d_z = -1.36$), and for DiCE, lower in 67 of 68 runs (median -0.210 , $d_z = -2.15$). For LIME the deficit was consistent, lower in 73 of 75 runs, but small in absolute terms (median -0.0034), because LIME’s stability on these data is close to zero on average. For Anchors the median was close to zero.

Fidelity moved in opposite directions. It was lower on errors for DiCE (median -0.053) and Anchors (-0.022), higher for LIME ($+0.032$), and not significantly different for SHAP after correction ($p_{\text{Holm}} = 0.057$). The faithfulness gap was lower on errors for SHAP (median -0.023 , lower in 55 of 65 runs) and DiCE, and did not differ for LIME and Anchors. Of the 12 primary tests, 9 were significant after correction and 3 met the practical-relevance criterion: SHAP stability, DiCE stability and DiCE fidelity. The robustness analysis over model–seed units agreed in direction and significance for 10 of the 12 tests; the exceptions were the LIME faithfulness gap, not significant in either analysis, and the DiCE faithfulness gap, which lost significance.

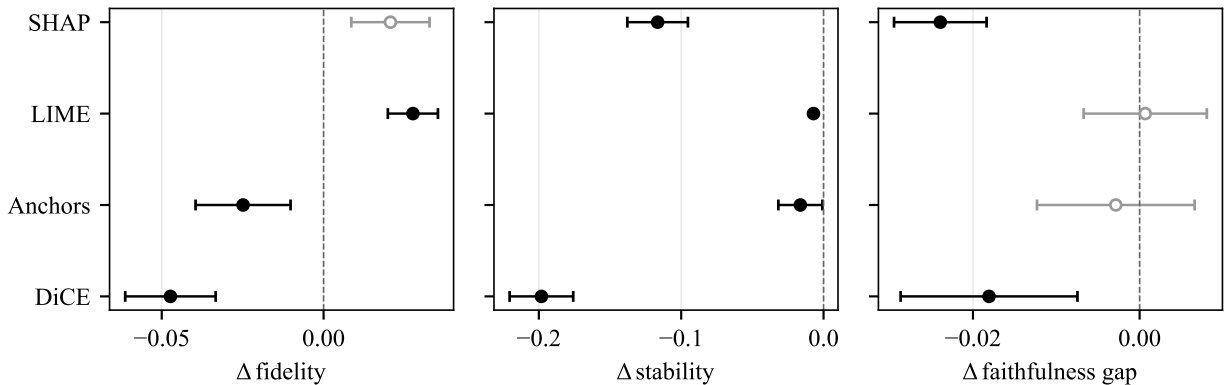


Figure 1: Mean per-run difference in quality between misclassified and correct instances, with 95% confidence intervals (UCI Adult). Filled markers: significant after Holm correction; open markers: not significant. Negative values mean lower quality on errors.

Sparsity differed little in absolute terms (all medians within ± 0.0025 of zero apart from Anchors). Explanation time did not differ systematically, except for Anchors, which took a median

Table 2: Per-run difference in quality, misclassified minus correct instances (UCI Adult). n : runs; CI: 95% confidence interval of the mean; p_{Holm} : Wilcoxon signed-rank, Holm-corrected within the 12 primary and, separately, the 4 sparsity tests.

Metric	Explainer	n	Median Δ	Mean Δ [95% CI]	d_z	p_{Holm}
Stability	SHAP	65	-0.076	-0.117 [-0.138, -0.095]	-1.36	<0.001
	LIME	75	-0.003	-0.007 [-0.009, -0.006]	-1.06	<0.001
	Anchors	57	0.000	-0.016 [-0.032, -0.001]	-0.28	<0.001
	DiCE	68	-0.210	-0.198 [-0.221, -0.176]	-2.15	<0.001
Fidelity	SHAP	65	+0.007	+0.021 [+0.009, +0.033]	+0.42	0.057
	LIME	75	+0.032	+0.028 [+0.020, +0.035]	+0.82	<0.001
	Anchors	57	-0.022	-0.025 [-0.040, -0.010]	-0.45	0.013
	DiCE	68	-0.053	-0.047 [-0.061, -0.033]	-0.82	<0.001
Faithfulness gap	SHAP	65	-0.023	-0.024 [-0.029, -0.018]	-1.07	<0.001
	LIME	75	0.000	+0.001 [-0.007, +0.008]	+0.02	0.992
	Anchors	57	0.000	-0.003 [-0.012, +0.007]	-0.08	0.954
	DiCE	68	-0.022	-0.018 [-0.029, -0.007]	-0.41	0.013
Sparsity	SHAP	65	0.000	+0.009 [+0.006, +0.012]	+0.85	<0.001
	LIME	75	+0.001	+0.002 [+0.001, +0.003]	+0.44	0.003
	Anchors	57	+0.005	+0.014 [+0.007, +0.020]	+0.58	<0.001
	DiCE	68	-0.003	-0.003 [-0.003, -0.002]	-1.18	<0.001

3,338 ms longer per instance on errors, plausibly because its rule search runs longer near the boundary.

3.2 Differences across model families (RQ2)

The size, and for fidelity and the faithfulness gap also the sign, of the difference depended on the model family (Figure 2). The Kruskal–Wallis test was significant after correction for 10 of the 12 explainer–metric combinations; the two exceptions were the stability of LIME and Anchors, whose differences were small for every family. The SHAP stability deficit was largest for the multilayer perceptron (median -0.216) and XGBoost (-0.168) and smallest for the random forest (-0.025) and the SVM (-0.030). The DiCE stability deficit was present for every family and largest for logistic regression (-0.321).

3.3 False positives versus false negatives (RQ3)

Among errors, the faithfulness gap was much larger for false positives than for false negatives for every explainer (Table 3); for SHAP and LIME this held in every run. DiCE stability was lower on false positives. The other differences were small. A likely reason for the faithfulness-gap asymmetry

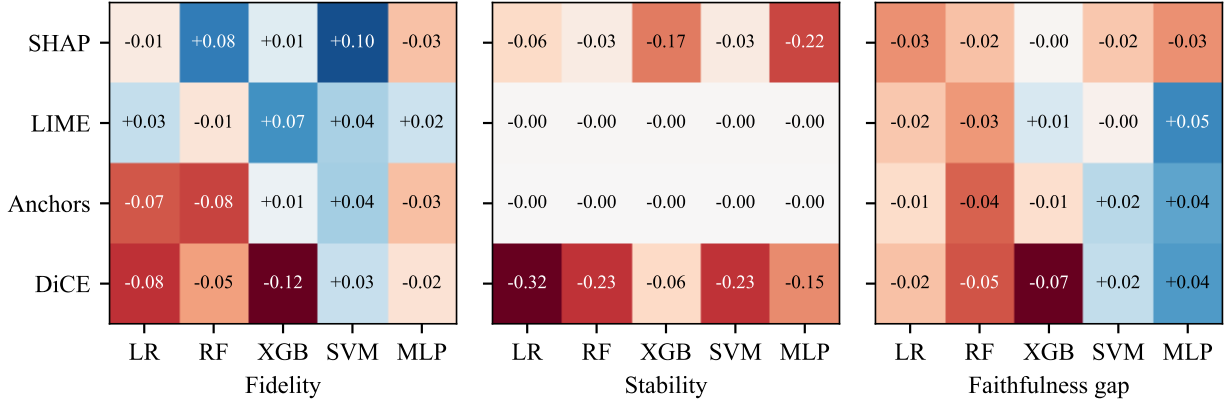


Figure 2: Median per-run difference (misclassified minus correct) by explainer and model family. LR: logistic regression; RF: random forest; XGB: XGBoost; SVM: support vector machine; MLP: multilayer perceptron. Red: lower quality on errors; blue: higher.

is how the metric masks features: masking to the training mean moves an instance towards the majority, negative class, so removing the top features changes a positive prediction (a false positive) far more than a negative one (a false negative). The gap is therefore not comparable across predicted classes, which matters for any audit that compares explanations of different decisions.

Table 3: Per-run difference, false positives minus false negatives: median Δ (p_{Holm} , Wilcoxon signed-rank, Holm-corrected across the 12 tests).

Explainer	Fidelity	Stability	Faithfulness gap
SHAP	+0.022 (<0.001)	+0.015 (0.705)	+0.353 (<0.001)
LIME	+0.003 (0.315)	+0.004 (<0.001)	+0.239 (<0.001)
Anchors	+0.027 (0.002)	0.000 (0.705)	+0.172 (<0.001)
DiCE	-0.019 (0.276)	-0.269 (<0.001)	+0.046 (<0.001)

3.4 The role of the decision boundary (RQ4)

Errors were indeed closer to the boundary (median margin 0.18 against 0.43 for correct instances in the SHAP runs), and stability rose with the margin for every explainer (Figure 3). Controlling for the margin changed the picture in three ways (Table 4). First, the DiCE stability deficit was almost entirely a margin effect: its coefficient fell from -0.191 to -0.014 , with a confidence interval including zero. A counterfactual for an instance near the boundary needs only a small change, and small changes vary more under noise. Second, the SHAP stability deficit was reduced but remained: -0.146 without and -0.042 $[-0.057, -0.027]$ with the margin. Figure 3 shows the same: at equal margin, SHAP explanations of errors are less stable than those of correct predictions. Third, the

faithfulness gap, which in RQ1 was lower on errors only for SHAP and DiCE and by small amounts, became lower on errors once the margin was held fixed for SHAP (-0.064), LIME (-0.031) and Anchors (-0.017). Near the boundary any deletion moves the prediction a lot, which inflates the gap of errors; at equal margin, the features named first carry less of an erroneous prediction. The fidelity differences were largely margin effects: the LIME advantage on errors fell from $+0.039$ to $+0.014$, the Anchors deficit vanished, and the DiCE deficit reversed sign.

Table 4: Coefficient of misclassification β (95% CI) in run fixed-effects regressions, without and with the prediction margin; standard errors clustered by run. Runs whose recorded predictions the stored model reproduces.

Metric	Explainer	β without margin	β with margin
Stability	SHAP	$-0.146 [-0.170, -0.123]$	$-0.042 [-0.057, -0.027]$
	LIME	$-0.008 [-0.009, -0.006]$	$-0.003 [-0.004, -0.002]$
	Anchors	$-0.030 [-0.062, +0.001]$	$-0.025 [-0.058, +0.008]$
	DiCE	$-0.191 [-0.219, -0.163]$	$-0.014 [-0.031, +0.004]$
Fidelity	SHAP	$-0.002 [-0.011, +0.007]$	$-0.004 [-0.011, +0.003]$
	LIME	$+0.039 [+0.032, +0.046]$	$+0.014 [+0.006, +0.021]$
	Anchors	$-0.013 [-0.032, +0.005]$	$+0.012 [-0.003, +0.027]$
	DiCE	$-0.045 [-0.062, -0.028]$	$+0.022 [+0.009, +0.034]$
Faithfulness gap	SHAP	$-0.024 [-0.031, -0.017]$	$-0.064 [-0.075, -0.054]$
	LIME	$+0.008 [0.000, +0.016]$	$-0.031 [-0.045, -0.018]$
	Anchors	$+0.003 [-0.006, +0.013]$	$-0.017 [-0.025, -0.008]$
	DiCE	$-0.013 [-0.025, 0.000]$	$-0.007 [-0.014, 0.000]$

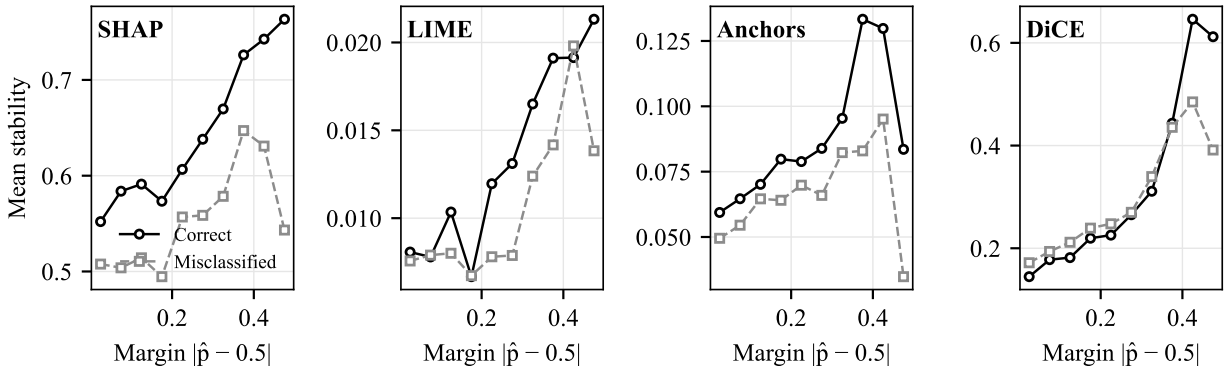


Figure 3: Mean stability by prediction margin $|\hat{p} - 0.5|$ for correct and misclassified instances (UCI Adult; bins of width 0.05 with at least 30 instances).

3.5 External check on German Credit

On German Credit, Anchors reproduced the Adult pattern: fidelity and faithfulness gap were lower on errors in all six runs (medians -0.084 and -0.017) and stability in five (-0.043). SHAP did not: its stability was higher on errors in 4 of six runs (median $+0.025$), while its fidelity was lower in all six (-0.045). With six runs these are descriptive. They show that the size, and for SHAP the direction, of the difference depends on the data as well as on the model.

4 Discussion

The answer to the title question is a qualified yes. On the Adult benchmark, explanations of wrong predictions were less stable for every explainer, and for SHAP and DiCE the difference was large: a SHAP explanation of a misclassified instance is markedly more sensitive to small input noise than one of a correct prediction. Averages over all instances hide this, because correct and incorrect predictions are pooled. Because the benchmark sampled the four quadrants in balance, the pooled average even overweights errors relative to their real frequency; in deployment, where errors are rarer, the average would hide the deficit further.

The margin control separates two explanations of the deficit. For DiCE the cause is geometric: errors are near the boundary, and explanations near the boundary are less stable whether or not the prediction is right. For SHAP a deficit remains at equal margin, so correctness itself, or something correlated with it beyond the margin, matters. The faithfulness gap shows the opposite masking effect: the raw contrast understates the deficit because near-boundary instances are easy to move. In both directions, comparing explanation quality across instances without accounting for the margin can mislead.

Three practical consequences follow. First, explanation benchmarks and audits should report quality separately for correct and misclassified instances, or at least by error quadrant, rather than only as an average. Second, when an explanation of an individual decision is used to contest or justify it, its stability should be checked for that instance, since it is lower precisely for the decisions that are most likely to be contested. Third, deletion-based metrics with a mean baseline are not comparable across predicted classes (RQ3) and should be reported per class.

The study has limits. The main dataset is a single tabular benchmark; the external check covers two explainers on one more dataset, with only six runs each, and it did not reproduce the SHAP stability deficit. The data are existing runs, so the analysis could not change the explainer settings; LIME's stability is near zero on these data, which limits what can be said about it. Anchors and DiCE have missing runs that are not missing at random, although the within-run design means missing runs remove units without biasing the contrast within the runs that exist. RQ4 could use

only part of the random-forest runs. The models were trained on the partition of one seed, so for the other seeds the sampled instances partly overlap the training data; this changes which instances are errors, not how the contrast within a run is computed, but errors on training instances may be atypical. The metrics are proxies: lower stability or faithfulness on errors does not by itself show that users would be misled, which requires a user study.

5 Conclusions

Post-hoc explanations were less reliable when the model was wrong. In a pre-specified analysis of 265 benchmark runs, stability was lower on misclassified instances for SHAP, LIME, Anchors and DiCE, with large deficits for SHAP and DiCE. The DiCE deficit is explained by the proximity of errors to the decision boundary; the SHAP deficit is not. At equal margin, the features that explanations rank first carry less of an erroneous prediction. The pattern depends on the model family and the dataset. Explanation quality should therefore be reported separately for correct and incorrect predictions, because an average describes the explanations that are needed least.

Data availability

Code, run artifacts and the analysis script (`scripts/pubs/paper_d_analysis.py`) are available at <https://github.com/jonnabio/xai-eval-framework>; an archived release will be cited.

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